
Dexterous Hand Flex2

Operation Manual

XynovaHandPilot Upper Computer Software

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Hangzhou Xynova Technology Co., Ltd.

Table of Contents

1. Precautions	4
1.1. Safety and Mechanics	4
1.2. Electrical and Communication	4
1.3. Software Operation	4
2. DISCLAIMER AND LIMITATION OF LIABILITY	6
3. Product Introduction	8
3.1. Product Overview	8
3.2. Communication Description	8
3.3. Software Capability Overview	9
4. Upper Computer Software Usage	10
4.1. System Installation and Startup	10
4.1.1. Recommended Operating Environment	10
4.1.2. Startup Steps	10
4.2. Interface Overview	10
4.3. Device Connection (CANFD)	12
4.3.1. Connection Steps	12
4.3.2. Left/Right Hand Identification and Data Segregation	12
4.3.3. Disconnection and Communication Loss	12
4.4. Console – Control	13
4.4.1. Joint Tree	13
4.4.2. 3D Model Interaction	13
4.4.3. Joint Adjustment Tool	14
4.4.4. Real-Time Data Card	14
4.5. Console – Motion	15
4.5.1. Default Gestures	15
4.5.2. Custom Gestures (Segregated by Left/Right Hand)	16
4.5.3. Glove Teleoperation Recording Playback	16
4.5.4. Motion Library Editing and Playback	16
4.6. Console – Glove Teleoperation	17
4.6.1. Connection and Mutual Exclusion	17
4.6.2. Recording Motions	17
4.7. Console – Vision Teleoperation	19
4.7.1. Operation Steps	19
4.8. Monitoring Station	20
4.8.1. Common Controls	20
4.8.2. Tactile Sensors	21
4.8.3. Force Analysis	21
4.8.4. Real-Time Temperature / Joint Overview / Joint Angles	22
4.9. op Bar Overlays: Wrist Camera and Proximity Sensor	23
4.9.1. Wrist Camera	23
4.9.2. Proximity Sensor	23
4.10. Device Maintenance	24
4.10.1. Device Self-Test	24
4.10.2. Fault Snapshot	25
4.10.3. Operation Log	26
4.11. Settings	26
4.11.1. Firmware Upgrade	27

4.12. Common Dialog Overview	28
5. Recommended Operation Procedures	29
5.1. Initial Power-On Commissioning	29
5.2. Daily Use	29
6. Common Troubleshooting	30
7. Appendix A: Local Data File Description	31
8. Appendix B: UI Screenshot Notes	31
9. Appendix C: Technical Support	31

1. Precautions

To ensure personnel safety and the reliable operation of the dexterous hand Flex2 and the host computer software XynovaHandPilot, please read this chapter carefully before powering on, wiring, installation, and software operation, and strictly comply with it during daily use.

1.1. Safety and Mechanics

- During power-on and communication, do not touch the moving fingers, wrist joints, or transmission mechanisms to prevent pinching injuries.
- The equipment must be securely mounted. Confirm that the mounting position possesses sufficient mechanical strength to support the motion load of the dexterous hand.
- Before enabling for the first time or issuing large-range motion commands, confirm that there are no personnel, obstacles, or tangled cables within the workspace.
- Do not forcibly manipulate the fingers or wrist while the joints are in motion.

1.2. Electrical and Communication

- Confirm that the supply voltage matches the product specifications. Inrush current may occur at the moment of power-on; the power supply must have sufficient margin and overcurrent protection.
- Communication employs CAN FD. Please wire according to the interface definition to avoid short circuits, reverse polarity, and loose connections.
- A $120\ \Omega$ termination resistor must be configured at the bus end as required by the system (if not integrated within the product, it must be matched on the user side).
- After communication disconnection, heartbeat timeout, or emergency stop trigger, troubleshoot the fault before reconnecting. Avoid repeated enabling with active faults.

1.3. Software Operation

- Glove teleoperation and vision teleoperation are mutually exclusive. Do not start them simultaneously.
- When recording/playing actions or loop-playing the action library, ensure that the area around the dexterous hand is safe.
- Upon disconnection, the software will reset the local posture and collapse the joint tree selection. Please be aware of this behavior.

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- Custom gestures and glove-recorded actions are stored separately for the left hand and right hand. The list will automatically switch when connecting to a different hand side.

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Hangzhou Xynova Technology Co., Ltd.

3. Product Introduction

3.1. Product Overview

The Dexterous Hand Flex2 is a multi-degree-of-freedom dexterous hand product designed for embodied intelligence and fine manipulation scenarios, supporting CANFD communication. The companion upper computer software, XynovaHandPilot, runs on the Windows platform and provides complete functionality including device connection, 3D model manipulation, gesture/motion orchestration, glove teleoperation, vision teleoperation, sensor monitoring, device self-test, and fault diagnosis.



Figure 1. Flex2 Dexterous Hand Illustration

3.2. Communication Description

Flex2 communicates with the upper computer via CANFD (commonly using the ZLG USBCANFD adapter). After a successful connection, the software reads the device identification information (model, serial number, firmware version, left/right hand identifier) and automatically loads the corresponding left/right hand 3D model as well as the local gesture and recording libraries.

3.3. Software Capability Overview

Module	Description
Console – Control	Joint tree selection, angle/deflection adjustment, 3D model interaction, save custom gestures
Console – Motion	Default gestures, custom gestures, motion library editing and loop playback, glove recording playback
Console – Glove Teleop	Connect hand/wrist, start/stop recording motions
Console – Vision Teleop	Camera preview, keypoint quality, start/stop vision teleoperation
Monitoring Station	Tactile sensing, force analysis, temperature, joint overview/angles; view by signal or by object
Device Maintenance	Device self-test, fault snapshot, operation log
Top Bar Overlays	Wrist camera, proximity sensor array display
Settings	Language, upgrade entry, fault popup, heartbeat monitoring, product SN, etc.

Table 1. XynovaHandPilot Main Functional Modules

4. Upper Computer Software Usage

XynovaHandPilot is a Windows-based upper computer application for the Dexterous Hand Flex2, designed for a resolution of 1920 x 1080 (minimum window 1280 x 720, maximized by default). The interface follows a three-tab navigation structure: Console / Monitoring Station / Device Maintenance, complemented by the top bar device connection and quick toggles, achieving a complete closed loop from connection to control, teleoperation, monitoring, and maintenance.

4.1. System Installation and Startup

4.1.1. Recommended Operating Environment

- Operating System: Windows 10/11 64-bit
- Display: Recommended 1920 x 1080 or higher
- Hardware: ZLG USBCANFD (or compatible CANFD adapter) and driver; optionally equipped with glove system, vision camera (RealSense D405 pipeline), wrist camera (Orbbec Gemini 305)
- Dependencies: Qt, ControlCANFD.dll, camera SDK, and other runtime libraries provided with the software package must be placed in the same directory as the main program or arranged per the deployment instructions

4.1.2. Startup Steps

1. Confirm that the dexterous hand power supply, CAN bus, and adapter are correctly connected, with the adapter plugged into the computer.
2. Extract/install the software package, and double-click to run XynovaHandPilot.exe (or the main program in the release package).
3. After the software launches maximized, the top bar displays the connection status (default: not connected), with the navigation bar on the left.

Note: Before first use, confirm that the hardware wiring harness, voltage, and communication settings are compatible. When the device is not connected, most motion and teleoperation functions are unavailable.

4.2. Interface Overview

The main interface consists of the top bar, left navigation, central content area, and various function panels. The top bar provides the brand logo, device connection status chip, wrist camera switch, proximity

sensor switch, fault badge, and window control buttons. The left navigation switches among Console, Monitoring Station, and Device Maintenance. The gear button at the bottom opens the Settings dialog.



Figure 2. Console Main Interface Example (Control)

Area	Name	Function Description
Top Bar	Connection Status / Device	The chip displays Connected / Not Connected; shows model and left/right hand; click to open the device popup
Top Bar	Wrist Camera	Opens the Orbbec wrist camera floating window (color/depth)
Top Bar	Proximity Sensor	Opens the 4×4 proximity sensor overlay (requires device connected)
Top Bar	Fault Badge	Displays fault count; click to jump to Maintenance – Fault Snapshot
Left Side	Console	Control / Motion / Glove Teleop. / Vision Teleop.
Left Side	Monitoring Station	Sensor and curve monitoring
Left Side	Device Maintenance	Self-Test / Fault Snapshot / Operation Log
Left Bottom	Settings	Language, upgrade, heartbeat monitoring, etc.

Table 2. Main Interface Area Description

4.3. Device Connection (CANFD)

4.3.1. Connection Steps

1. Connect the USBCANFD to the computer. Power on the dexterous hand and connect it to the CANFD bus.
2. Click the top bar device chip or "Select Device" to open the device popup.
3. Select the correct adapter/interface from the candidate list (e.g., USBCANFD, channel number).
4. Click "Connect". The button briefly shows "Connecting...". Upon success, the status changes to "Connected" and displays the model, left/right hand, SN, and firmware version.
5. Upon successful connection, the software automatically: loads the left/right hand 3D model; switches the custom gesture/glove recording library to the data corresponding to the hand side; and performs a background pre-check when necessary.

4.3.2. Left/Right Hand Identification and Data Segregation

The hardware identifies whether the hand is left or right. After connection, the software automatically recognizes this and reads/writes separately:

- Custom gestures: `custom_gestures_left.json` / `custom_gestures_right.json`
- Glove teleoperation recordings: `glove_recordings_left.json` / `glove_recordings_right.json`

The storage path is located in the application data folder under the Windows user directory (AppData\Xynova\XynovaHandPilot).

When connected to the left hand, only left-hand data is displayed/saved; when connected to the right hand, only right-hand data is displayed/saved.

4.3.3. Disconnection and Communication Loss

- Manual Disconnection: Click "Disconnect" in the device popup. The software will stop teleoperation/playback, reset joints to zero, clear the information card, collapse the joint tree and deselect, and navigate back to Console – Control.
- Heartbeat Timeout: If "Heartbeat Monitoring" is enabled in Settings, a prompt will appear asking whether to reconnect upon communication loss. You may choose to reconnect.

Note: After communication disconnection, all joint tree dropdown menus will collapse and joint selection states will be cleared.

4.4. Console – Control

"Control" is the default mode of the Console and the only mode that allows drag-to-rotate/pan the view on the 3D model and directly adjust joints. Glove teleoperation, vision teleoperation, monitoring, and maintenance pages lock the model drag functionality to avoid conflicts with teleoperation commands.



Figure 3. Control Mode: Joint Tree + 3D + Joint Adjustment Tool

4.4.1. Joint Tree

The "Joint Tree" on the left is grouped by finger: Thumb, Index Finger, Middle Finger, Ring Finger, Little Finger, Wrist. Click a group to expand/collapse; when expanded, the joints of that group are shown (e.g., TH1..., FF1...FF4, WR1/WR2, etc.). Click a joint name to select it (single selection). Click the same group again to collapse.

- Full panel collapse (collapse/expand joint tree) is supported to enlarge the central 3D display area.
- After communication disconnection, all groups collapse and all selections are cleared.

4.4.2. 3D Model Interaction

The center displays a WebEngine-based URDF 3D hand model. After connecting and identifying the hand side, the left or right hand model is loaded. In Control mode, you can:

- Click a model joint: linked selection with the joint tree
- Drag empty space: rotate/pan the view (locked in non-Control modes)
- Work with the joint adjustment tool to drive target angles in real time and synchronously send

4.4.3. Joint Adjustment Tool



Figure 4. Real-Time Data Card and Adjustment Area After Selecting a Joint

After selecting a joint, the bottom/floating "Joint Adjustment" tool becomes available:

- Angle: Displays the current target angle; increase/decrease via +/- point press or long press; numeric value can also be clicked and edited (constrained by joint limit range).
- Deflection: Provides deflection adjustment for WR1/WR2 and some distal joints (e.g., FF4/MF4/RF4/LF4).
- Reset Position: Returns all joint target angles to zero.
- Save Gesture: Saves the current whole-hand posture as a custom gesture (opens naming dialog).

4.4.4. Real-Time Data Card

After selecting a joint, an information card may appear, displaying real-time curves and key data, supporting pin, freeze/unfreeze, and zoom-in viewing.



Figure 5. Joint Real-Time Data Enlarged View

4.5. Console – Motion

The Motion mode is used to quickly execute default gestures, manage custom gestures, orchestrate the motion library, and playback glove teleoperation recorded motions.



Figure 6. Motion Panel: Default Gestures / Custom Gestures / Glove Recorded Motions

4.5.1. Default Gestures

The software includes commonly used built-in default gestures (examples): Finger Extension, Finger Flexion, Index

Finger Single Flexion/Extension, Thumb Opposition, Cylindrical Grasp, Spherical Grasp, Two-Finger Pinch, Three-Fingertip Pinch, Hook Grasp, Writing Gesture, Push-Press Grasp, etc. Click the card to execute; clicking again cancels and returns to the initial posture.

4.5.2. Custom Gestures (Segregated by Left/Right Hand)

1. Adjust to the desired posture in "Control" mode.
2. Click "Save Gesture", enter a name, and confirm.
3. The gesture appears in the "Custom Gestures" list. Click to reproduce.
4. "Select to Delete" enters deletion mode; click a card to delete a single item. "Clear Custom" clears all custom gestures for the current hand side.

Note: Custom gestures are saved separately for the currently connected left/right hand. When switching connection to the other hand, the list automatically switches to the corresponding hand-side data.

4.5.3. Glove Teleoperation Recording Playback

Motions recorded and saved during glove teleoperation appear in the "Glove Teleoperation Recorded Motions" list (also segregated by left/right hand). Click once to begin playback following the sampling timeline; clicking again stops playback and resets.

4.5.4. Motion Library Editing and Playback



Figure 7. Motion Library Editing Interface

Click "Motion Editor" to open the motion library:

- Create / Copy / Delete motions; Import / Export JSON.
- Drag gestures from the left into the motion sequence area, and set per-step delay (ms).
- Supports rename, save, and end editing; if unsaved changes exist, exiting prompts for confirmation.

The bottom transport bar allows selecting a motion, setting the loop count, and performing play/pause/stop. During playback, gestures are executed sequentially with the configured delay per step.

4.6. Console – Glove Teleoperation



Figure 8. Glove Teleoperation Page

Glove teleoperation is used to map the posture of an external glove (such as the Manus pipeline) to Flex2. Before use, first connect the dexterous hand and confirm that the glove software/driver is ready.

4.6.1. Connection and Mutual Exclusion

1. Go to Console → Glove Teleoperation.
2. Click "Connect Hand" and wait for the status to become online.
3. If wrist following/locking is needed, click "Connect Wrist" (usually requires hand already connected).
4. If vision teleoperation is currently active, stop vision teleoperation first.

4.6.2. Recording Motions

1. After the hand is online, click "Start Recording Motion". The button shows "Recording".

2. Perform the target motion sequence (~1 ms sampling by default, up to ~3 minutes maximum).
3. Click again to stop recording. Name and save in the "Save Recorded Motion" dialog.
4. After saving, the motion can be played back in the Motion Panel under "Glove Teleoperation Recorded Motions". The data is written to the corresponding hand-side JSON file.



Figure 9. Glove Teleoperation Related Overlay/Status Illustration

Note: During recording, do not switch hand sides or disconnect the device. Recording, playback, and vision teleoperation have mutual exclusion prompts; please follow the on-screen Toast guidance.

4.7. Console – Vision Teleoperation



Figure 10. Vision Teleoperation Page (Camera Preview and Keypoints)

Vision teleoperation estimates human hand keypoints through an external vision pipeline (commonly RealSense D405 + Desktop vision project) and maps them to control the dexterous hand. The page displays camera preview and keypoint quality (e.g., keypoints N/23).

4.7.1. Operation Steps

1. Confirm the dexterous hand is connected. If glove teleoperation is active, first disconnect/stop glove teleoperation.
2. Confirm that the vision runtime environment and camera are ready (start dependent services per the deployment package instructions).
3. Go to Console → Vision Teleoperation to view the preview and keypoint count.
4. Click "Camera Settings" to view read-only camera information (device, resolution, frame rate, hand side, etc.).
5. Click "Start Teleoperation" to enter the closed loop; when finished, click "Stop Teleoperation".

Note: Tracking is unstable when keypoints are too few. Ensure the hand is within the camera field of view, lighting is sufficient, and background interference is minimal.

4.8. Monitoring Station

The Monitoring Station is used to observe tactile sensing, force, temperature, and joint status, supporting two browsing dimensions (by signal / by object) and the ability to switch between the 3D hand model and the joint reference diagram.



Figure 11. Monitoring Station Main Interface

4.8.1. Common Controls

- Joint Reference Diagram Switch: When off, displays the 3D hand; when on, displays the left/right hand SVG joint legend.
- Data Freeze / Click to Unfreeze: Freezes curve and table refresh for convenient screenshot analysis.

4.8.2. Tactile Sensors



Figure 12. Monitoring Station – Tactile Sensors

Displays tactile unit (taxel) distribution and values by finger region (fingertip/finger pad/finger root, etc.) and palm, facilitating observation of contact areas.

4.8.3. Force Analysis



Figure 13. Monitoring Station – Force Analysis

The regional force analysis table displays $F_x/F_y/F_z$ and resultant force for each region, and can overlay force vectors on the 3D model to aid in determining grasp force direction.

4.8.4. Real-Time Temperature / Joint Overview / Joint Angles



Figure 14. Monitoring Station – Signal Curve Illustration



Figure 15. Monitoring Station – View Indicators by Joint

View the temperature curves of the main control board / signal board, as well as joint target angles, actual angles, fault status, etc. In "By Object" mode, you can focus on a single joint. Double-clicking a finger in the Console also jumps to the corresponding object in the Monitoring Station.

4.9. op Bar Overlays: Wrist Camera and Proximity Sensor

4.9.1. Wrist Camera

Toggling the top bar "Wrist Camera" switch opens a floating window displaying color and depth images, supporting screenshot capture and full screen. Screenshots can be saved as `wrist_color_*.png` / `wrist_depth_*.png`.

4.9.2. Proximity Sensor



Figure 16. Proximity Sensor Overlay

Toggling the "Proximity Sensor" switch displays a 4×4 distance array. When the device is not connected, a prompt to connect first is shown. The panel supports freeze display.

4.10. Device Maintenance

4.10.1. Device Self-Test



Figure 17. Device Maintenance – Device Self-Test

With the device securely mounted and the surrounding area safe, click "Start Self-Test". The interface will check items including CANFD communication, signal processing board, main control board, power supply, tactile module, parameter storage, firmware upgrade status, emergency stop status, etc., and display results as red/green result rows.

After a successful connection, the software may also automatically perform a background pre-check.

Note: The estimated time displayed in the interface text may differ from the actual software pre-check implementation; please follow the actual prompts and results. Full hardware long-duration self-test should follow on-site process specifications.

4.10.2. Fault Snapshot



Figure 18. Device Maintenance – Fault Snapshot

Displays the current fault list and details (operation commands before and after the fault occurrence, feedback waveform recording, suggested handling methods, etc.). Available actions:

- One-Click Fault Clear: Sends a fault clear command to the device (requires connected).
- Export Diagnostic Package: Exports logs and related information for after-sales analysis.

If "Popup Fault Info" is enabled in Settings, a "Deal Later / Immediate Maintenance" dialog pops up when a fault occurs. Clicking the top bar fault badge also navigates to this page.

4.10.3. Operation Log



Figure 19. Device Maintenance – Operation Log

The operation timeline records connection, motion, fault, system, and other events. Supports filtering by date, joint, signal, and level, as well as resetting filters and exporting logs.

4.11. Settings



Figure 20. Settings Dialog

Setting Item	Description
Language	Chinese / English (interface text subject to the actual release version)
Upgrade	View current version; select firmware type (Main Control Board / Sensor Board / Driver Board) and local upgrade entry
Popup Fault Info	When enabled, a modal prompt appears upon fault occurrence
Heartbeat Monitoring	When enabled, monitors communication heartbeat; prompts reconnection upon timeout (recommended to keep enabled)
Product S/N Code	Displayed after connection, read-only
Closed-Loop Mode	Interface locked as enabled, ensuring position closed-loop control

Table 3. Settings Item Description

After modification, click "Save" to apply changes; "Cancel" to discard changes.

4.11.1. Firmware Upgrade

The "Upgrade" area in the Settings dialog is used to perform online firmware upgrades for each module of the dexterous hand via CAN FD. During the upgrade, the software temporarily disables heartbeat monitoring and business communication (PDM/ODM); please ensure the device is connected and the power supply is stable. Do not power off or unplug the CAN FD adapter before the upgrade is complete.

Operation Steps:

1. Confirm the device is connected in the top bar, click the gear at the bottom left to open the Settings dialog, and enter the "Upgrade" area.
2. In the "Firmware Type" dropdown, select the target module (Main Control Board, Signal Processing Board, or FF1/FF3, MF1/MF3, RF1/RF3, LF1/LF3, TH1/TH2/TH3, WR1/WR2, FF4/MF4/RF4/LF4/LF5/TH4, etc.), and click "OK" to read and display the current version number.
3. Click "Upgrade", verify the target module in the confirmation box, then continue; follow the prompt to select the corresponding encoded .s19 firmware file (e.g., updata_encode1 / updata_encode2).
4. The software sends the upgrade data line by line according to the firmware file, and the progress bar displays the transfer progress in real time; to abort, click "Stop Upgrade".
5. When data transfer is nearly complete, the prompt "Data transfer complete, waiting for dexterous hand to upgrade" appears; the software then automatically determines the result and prompts success or failure.
6. After the upgrade process ends, business communication is automatically restored; if "Heartbeat Monitoring" is enabled in Settings, heartbeat monitoring will be restarted.

Notes:

Please confirm that the firmware file matches the selected firmware type and device model. If the model embedded in the file does not match the current device, the software will pop up a "Continue Upgrade Prompt", and the user must confirm whether to continue.

If no progress update occurs for approximately 5 seconds during transfer, the software determines a timeout and cancels the current upgrade; please check the connection and power supply, then retry.

When "System Upgrade Failed" or "Upgrade Version Mismatch" appears, please replace the file with the correct firmware package or contact the manufacturer for confirmation; avoid repeatedly upgrading with the wrong file.

Upgrades are supported only for connected devices; the "OK/Upgrade" buttons are unavailable when no device is connected.

4.12. Common Dialog Overview

Dialog	Purpose	Typical Buttons
Save Gesture	Name custom gesture	Cancel / OK
Save Recorded Motion	Name glove recording	Cancel / OK
Clear Custom	Clear all custom gestures for the current hand side	Cancel / OK
Rename Motion	Rename in motion library	Cancel / OK
Unsaved Changes – Confirm Exit	Dirty exit from motion editor	Cancel / OK
Camera Settings	Vision camera read-only information	OK
Communication Lost	Heartbeat lost – reconnect?	Yes / No
Fault Prompt	Deal later or immediate maintenance	Deal Later / Immediate Maintenance

Table 4. Common Dialogs

5. Recommended Operation Procedures

5.1. Initial Power-On Commissioning

1. Secure the dexterous hand and check the power supply and CANFD wiring.
2. Start XynovaHandPilot, connect the device, and confirm that left/right hand identification is correct.
3. In Control mode, use the joint tree to adjust a single joint within a small range and confirm that the motion direction and limits are normal.
4. Execute the default Finger Extension / Finger Flexion gestures to confirm full hand response.
5. Open the Monitoring Station to check whether tactile sensing / temperature / joint angles are refreshing.
6. If teleoperation is needed, verify the glove or vision link separately.

5.2. Daily Use

1. Connect device → complete tasks in Control or Motion mode.
2. When reproducing complex sequences is needed, use the motion library or glove recording playback.
3. If abnormalities occur, check the fault badge / fault snapshot, and export a diagnostic package if necessary.
4. When finished, disconnect, close the software, and finally cut off device power (per on-site specifications).

6. Common Troubleshooting

Symptom	Possible Cause	Suggested Action
CAN device not detected in scan	Driver not installed / loose cable / port occupied	Check adapter driver and port usage; re-plug and refresh the candidate list
Connection failure or frequent disconnection	Baud rate / termination resistor / unstable power / heartbeat issue	Check wiring and power; enable heartbeat monitoring for observation; check the operation log
3D model not displayed	Missing left/right hand model resources or service failure	Confirm installation includes 3DModelLeft/Right; check Toast prompts
Custom gesture list is empty or has changed	Connected to the other hand	Left/right hand data segregation is normal behavior; switch back to the corresponding hand side
Glove cannot connect	Dexterous hand not connected / vision in use / glove service not started	Connect hand first; stop vision; check glove runtime environment
Vision teleoperation has no keypoints	Camera / field of view / pipeline not started	Check camera and vision project; ensure hand is within the frame
Monitoring has no data	Device not connected or data is frozen	Connect the device; turn off data freeze
Fault cannot be cleared	Fault still exists / not connected	Resolve the hardware fault first, then use One-Click Fault Clear; export diagnostic package if necessary. Contact support.

Table 5. Common Issues and Suggested Actions

7. Appendix A: Local Data File Description

File	Description
custom_gestures_left.json	Left hand custom gestures
custom_gestures_right.json	Right hand custom gestures
glove_recordings_left.json	Left hand glove recording motions
glove_recordings_right.json	Right hand glove recording motions
action_library.json	Motion library (gesture name sequences)
opLog, etc.	Operation logs and other runtime data (subject to implementation)

Table 6. Local Persistent Files

When upgrading from an older single-file version, the software automatically migrates and copies data to left/right hand files to avoid data loss.

8. Appendix B: UI Screenshot Notes

All illustrations in this manual are taken from actual XynovaHandPilot software interface screenshots, for easy cross-reference during operation. If the interface details of your software version have been iteratively updated, please refer to the actual software interface.

9. Appendix C: Technical Support

For hardware specifications, interface definitions, detailed communication protocols, or after-sales support, please contact Hangzhou Xynova Technology Co., Ltd. Please provide as much of the following information as possible: software version, device SN, firmware version, left/right hand, fault screenshots / diagnostic package, and reproduction steps.